

MONTROSE 11 ENE, CO, USA

WMO: 745220

Lat: 38.5440N

Lon: 107.6928W

Elev: 2561

StdP: 74.11

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03060

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-15.0	-21.8	0.7	-12.1	-19.8	0.9	-10.3	5.9	-0.8	4.9	-2.5	1.3	N/A	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	27.1	11.2	25.9	11.0	24.7	10.8	13.8	21.4	13.1	21.2	12.6	20.8	2.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	11.7	15.1	10.6	10.9	14.9	9.7	10.3	14.6	47.6	21.5	45.6	21.3	43.8	21.0	16.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.1	4.3	3.7	DB	-21.0	29.2	2.6	0.8	-22.9	29.8	-24.5	30.3	-25.9	30.8	-27.8	31.4	(4)
			WB	-21.0	14.8	3.2	0.8	-23.3	15.4	-25.2	15.9	-27.0	16.4	-29.4	17.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-5.1	-3.7	0.4	4.2	9.1	15.7	18.5	17.1	13.3	6.4	0.8	-4.9
	DBStd	9.32	4.30	4.58	4.58	4.59	4.41	3.55	2.03	2.14	3.67	4.70	4.98	5.00
	HDD10.0	2278	468	385	296	177	70	5	0	0	15	125	276	461
	HDD18.3	4536	727	618	554	423	286	89	22	49	152	370	526	719
	CDD10.0	834	0	0	0	4	42	176	265	220	114	13	0	0
	CDD18.3	50	0	0	0	0	0	9	28	11	2	0	0	0
	CDH23.3	581	0	0	0	0	2	133	282	137	27	0	0	0
	CDH26.7	48	0	0	0	0	0	10	30	7	2	0	0	0
Wind	WSAvg	1.8	1.7	1.8	2.0	2.1	2.0	2.0	1.7	1.7	1.7	1.6	1.6	1.6
Precipitation	PrecAvg	356	27	29	30	33	29	16	27	32	41	38	28	27
	PrecMax	468	59	66	73	94	79	41	60	54	77	113	49	73
	PrecMin	255	5	6	1	13	1	0	3	11	10	7	7	15
	PrecStd	57	13	14	16	17	22	11	17	13	21	22	11	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	9.1	15.2	18.5	23.0	28.2	28.9	27.7	26.1	20.0	14.6	8.4
		MCWB	0.3	1.6	4.5	5.8	8.0	11.0	12.0	10.9	10.9	7.3	4.2	0.8
	2%	DB	4.5	6.7	12.4	16.2	20.7	26.2	27.5	26.3	23.8	17.7	12.3	5.6
		MCWB	-1.0	0.5	3.1	4.8	7.3	10.4	11.7	10.8	9.8	6.4	3.5	-0.4
	5%	DB	2.6	5.0	9.9	14.3	19.1	24.9	26.3	25.0	22.2	15.9	10.3	3.8
		MCWB	-1.9	-0.5	1.8	3.8	6.6	9.9	11.7	10.8	9.1	5.7	2.4	-1.0
	10%	DB	1.1	2.9	7.8	12.3	17.4	23.6	25.0	23.6	20.6	13.9	8.3	2.2
		MCWB	-2.6	-1.6	0.7	3.2	6.0	9.4	11.6	10.9	8.5	4.8	1.3	-1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	2.5	4.9	6.6	9.1	13.2	14.9	14.4	12.8	9.3	5.2	1.8
		MCDB	5.1	7.9	14.5	16.5	19.2	24.0	22.1	20.2	21.3	15.3	12.1	5.9
	2%	WB	0.1	1.2	3.7	5.5	7.9	12.0	14.1	13.6	11.7	8.0	3.9	0.8
		MCDB	3.2	6.1	12.0	14.2	18.1	23.2	21.7	20.6	19.7	15.2	10.8	4.1
	5%	WB	-0.9	0.2	2.4	4.6	7.0	11.2	13.5	12.9	10.9	6.8	2.6	-0.1
		MCDB	2.0	4.6	9.6	12.9	17.2	22.4	21.4	20.7	18.6	14.2	9.1	3.2
	10%	WB	-2.0	-0.6	1.3	3.6	6.3	10.3	12.9	12.3	10.1	5.9	1.6	-1.1
		MCDB	0.9	3.3	7.7	11.3	16.2	21.4	21.0	20.2	17.8	12.9	7.4	2.4
Mean Daily Temperature Range	5% DB	MDBR	8.7	8.4	9.7	11.0	11.7	13.1	12.8	12.4	12.3	10.8	9.9	8.1
		MCDBR	10.6	10.4	11.8	13.6	13.7	14.4	14.2	14.2	14.1	13.2	12.2	9.2
		MCWBR	7.1	6.0	5.8	6.4	6.1	5.9	5.0	4.9	5.4	6.1	6.3	5.5
	5% WB	MCDBR	9.3	9.1	11.4	12.4	13.2	13.2	11.8	12.0	11.7	11.5	11.5	8.2
		MCWBR	6.3	5.4	5.7	6.0	6.0	5.5	4.5	4.6	4.6	5.4	6.1	5.0
Clear-Sky Solar Irradiance	taub	0.216	0.220	0.245	0.264	0.276	0.275	0.312	0.308	0.271	0.249	0.225	0.217	
	taud	2.636	2.651	2.595	2.570	2.589	2.592	2.516	2.519	2.650	2.682	2.685	2.625	
	Ebn at Noon	1009	1041	1031	1018	1002	999	958	958	987	990	989	985	
	Edh at Noon	66	75	88	97	97	97	104	101	83	72	62	62	
All-Sky Solar Radiation	RadAvg	2.62	3.41	4.83	5.90	6.74	7.81	6.90	6.06	5.35	4.01	2.88	2.23	
	RadStd	0.17	0.24	0.39	0.36	0.53	0.49	0.40	0.28	0.30	0.33	0.21	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page